

VEILLENS: Explainable and Granular Analysis and Protection Against Visual Privacy Risks Beyond Visible Appearance

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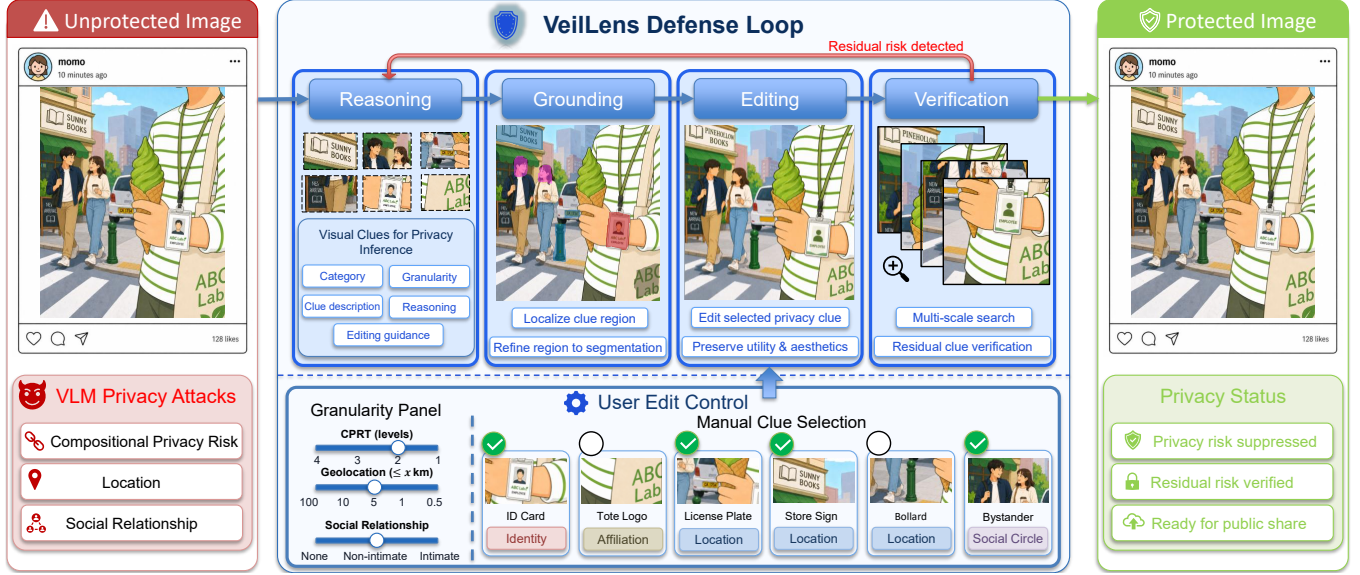


Figure 1: Given an unprotected image and user-specified protection preferences, VEILLENS identifies visual clues supporting privacy attacks. It then localizes clues, edits user-selected evidence, and verifies residual leakage through an iterative defense loop, producing a protected image that suppresses sensitive inference while remaining suitable for public sharing.

Abstract

Recent vision-language models (VLMs) have substantially elevated the threat level of visual privacy. A single shared image can reveal sensitive information not only through explicit identifiers, such as faces, text, or license plates, but also through weak and compositional cues that enable semantic reasoning about identity, precise locations, and social relationships. Existing visual privacy tools primarily target predefined visible identifiers or apply generic image transformations, making them ill-suited for reasoning-capable VLM attackers and offering limited explainability, granularity, and user control. We present VEILLENS, an explainable framework for analyzing and mitigating VLM-based visual privacy risks beyond explicit visual identifiers. Given an image, VEILLENS identifies privacy-relevant visual cues at multiple granularities, grounds them in the image, edits them through controllable generative transformations, and verifies whether the protected image still supports sensitive inference. This reasoning-grounding-editing-verification loop exposes actionable privacy risks while progressively reducing residual leakage after each protection round. We evaluate VEILLENS on

nearly 1,500 images with more than 20,000 VLM API calls, covering compositional privacy risk, geolocation leakage, and social-relationship inference. Across 500 compositional-privacy images, VEILLENS reduces the mean model-inferred privacy risk score by 48.7%. For geolocation privacy, it reduces block-level (≤ 0.5 km) localization success from 42.0% to 12.2% under Gemini-3 Flash. A user study with 25 participants further shows that VEILLENS improves users’ privacy awareness while preserving image utility, with 70.7% of responses preferring the protected image.

1 Introduction

Photo sharing has become a routine global practice: billions of images are uploaded each day to platforms such as Instagram, X, and Facebook [3]. Yet these shared images can expose individuals to privacy risks that extend far beyond explicit identifiers such as faces and license plates [2]. A single photograph can reveal geographic location, daily routines, social relationships, and institutional affiliations through visual cues that users may neither intend to disclose nor recognize as sensitive [3, 12, 51].

Unfortunately, the rapid deployment of frontier VLMs into everyday applications has fundamentally altered this threat landscape [8–10, 29, 31]. The advanced reasoning capabilities that make these models useful also enable powerful privacy inference: a low-effort adversary can upload a victim’s photo and obtain detailed inferences that previously required expertise or specialized pipelines.

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Recent work shows that VLMs can infer precise locations from weak visual signals through compositional reasoning [23], and recover identity or sensitive attributes from contextual cues that users may neither recognize nor aggregate [20, 43]. *As a result, visual privacy protection must move beyond detecting and obscuring predefined visible identifiers to also account for compositional cues, helping users understand, control, and mitigate the broader privacy risks exposed by ordinary shared images.*

Existing visual privacy defenses fall short along three axes. First, **scope**: many target specific risks such as faces, license plates, text, or geolocation [43], limiting their ability to address inference from general visual context. Second, **threat model**: many are evaluated against human observers or narrow CV classifiers [1, 28], rather than reasoning-capable VLM adversaries. Third, **verifiability**: existing defenses rarely verify whether the protected image still leaks the target private attribute [4], leaving users with little assurance after protection.

To address these limitations, we propose VEILLENS, an explainable and user-controllable visual privacy protection framework against reasoning-capable VLM adversaries. As shown in Fig. 3, rather than treating privacy protection as a one-shot sanitization problem, VEILLENS formulates it as a *reasoning-grounding-editing-verification* loop: given an input image and a user-specified privacy objective, VEILLENS uses a trusted VLM oracle to reason about what sensitive information may be inferred and why, grounds the supporting visual evidence as editable regions, applies user-controllable generative inpainting to selected clues, and iteratively verifies whether the protected image still leaks the target attribute.

Building VEILLENS introduces three key technical challenges.

First, how can VEILLENS accurately reason about and ground potential privacy leakage while remaining explainable to users? Although a VLM oracle can identify privacy risks at a high level, it often does not localize the specific visual evidence responsible, limiting transparency and user trust. To address this, VEILLENS decouples reasoning from localization: a VLM identifies privacy-relevant cues in language, which are then grounded into pixel-level segmentation masks through an open-set detection-and-segmentation pipeline, exposing the evidence chain to the user.

Second, how can VEILLENS enable targeted privacy editing while preserving image utility and keeping users in control? Different users have different privacy protection objectives: some prioritize location suppression, others identity or institutional affiliation. Thus, applying indiscriminate redaction would overwhelm users and unnecessarily degrade image quality. To address this, VEILLENS overlays each inferred cue with its associated privacy risk for user-controlled selection, then applies diffusion-based inpainting conditioned on the selected cue, achieving targeted removal while preserving perceptual realism.

Third, how can VEILLENS verify protection against strong VLM-based adversaries? A single pass of reasoning and editing may not eliminate all leakage: a VLM oracle tends to surface only the most salient cues, while a VLM adversary can still exploit residual contextual signals to recover the target attribute. To address this, VEILLENS introduces an iterative verification loop: after each editing round, an oracle VLM re-evaluates the protected image and reports residual evidence. VEILLENS grounds the reported cues and performs

additional inpainting until the target inference is suppressed or a user-defined budget is exhausted.

We implement VEILLENS as an end-to-end pipeline integrating a frontier VLM oracle [10, 30], an open-set detector and segmentation model, and a diffusion-based inpainting backend. We evaluate VEILLENS at scale across nearly 1,500 images and more than 20,000 VLM API calls, covering four frontier VLM adversaries, four recent defense baselines, and three privacy settings. For general compositional privacy, VEILLENS reduces the mean CPRT privacy risk score from 0.314 to 0.161, a 48.7% relative reduction. For geolocation, VEILLENS reduces block-level localization success from 42.0% to 12.2% under Gemini-3F, and achieves a 2.03 $\times$ lower attack success rate than the strongest prior defense [22]. For social-relationship privacy, VEILLENS reduces VLM relationship inference accuracy by over 92% across four frontier attackers [16]. A user study with $N = 25$ participants further shows the privacy and utility trade-off of VEILLENS. 70.7% (106/150) of responses prefer the protected image, and 52% (13/25) report willingness to use VEILLENS in everyday sharing today.

Our key contributions are as follows:

- We design an explainable and user-controllable visual privacy protection framework that exposes privacy-relevant visual evidence to users and applies privacy-conditioned generative inpainting to selectively remove chosen clues while preserving image utility.
- We introduce an iterative verification loop using a VLM oracle to achieve better privacy guarantee, which verifies protected images after editing and drives further targeted inpainting when residual leakage remains.
- We conduct a comprehensive evaluation of VEILLENS on nearly 1,500 images using more than 20,000 VLM API calls, four frontier VLM adversaries, four recent baselines, and three privacy settings, demonstrating that VEILLENS improves privacy protection and user awareness while preserving a favorable privacy-utility trade-off.

2 Related Work

2.1 Visual Privacy Leakage Beyond Explicit Visible Cues

Image sharing has become ubiquitous, yet users often lack awareness of privacy risks that shared images can expose [5]. Recent work shows that visual privacy cannot be reduced to a few explicit identifiers. Instead, many privacy risks arise through *compositional inference*, where multiple weak or contextual cues jointly reveal a sensitive attribute that is not directly visible [43]. Advances in VLMs further amplify this risk by enabling reasoning beyond visible markers toward richer, context-dependent privacy inferences [42]. Human-centered studies likewise suggest that privacy perception depends not only on individual objects, but also on co-presence, capture context, and higher-level reasoning [49]. Prior image privacy classifiers typically predict a binary public/private label from predefined visual entities such as scene and object types [48], but such fixed-category approaches are insufficient when leakage arises from compositional reasoning over contextual evidence. Motivated by this shift, VEILLENS is designed as a general framework to analyze and mitigate privacy risks that emerge beyond explicit visible cues. VEILLENS focuses on the following representative privacy

Method	Visual privacy threat	Defense target	Reasoning-aware defense	Privacy granularity	Clue explainability	User steering	Utility preserving	Verification loop
MM-PGD [15]	Geolocation	CV models						
ADGAN [47]	Geolocation	CV models					✓	
LocGuard [25]	Geolocation	CV models						
GeoShield [22]	Geolocation	VLM	✓		✓			
SVIA [19]	PII	CV models					✓	
Redaction [32]	PII	Human, OCR		✓			✓	
Cardea [40]	PII	Human, CV		✓	✓	✓		
Aragorn [44]	PII	Human, CV		✓	✓	✓		
CAIAMAR [4]	PII	General	✓		✓	✓	✓	✓
HR-VISP [1]	PII, Social relationship	CV models		✓	✓			
Imago Obscura [28]	General visual cues	Human	✓	✓	✓	✓	✓	
VEILLENS (Ours)	General (Geo, PII, Social relationship)	General (Human, VLM, CV)	✓	✓	✓	✓	✓	✓

Table 1: Comparison of prior visual privacy defenses and VEILLENS across protection scope, defense target, and key capabilities.

threats, while keeping extensible to additional emerging privacy risks.

Visual geolocation inference attacks. Visual geolocation inference attacks seek to infer an image’s capture location from its visual content and semantics. Early work formulates this problem as *visual place recognition* in classification or retrieval settings, matching a query image against large-scale geo-tagged databases or discretized geographic grids [39]. More recent VLM-based attacks reframe geolocation as a reasoning problem over weak and heterogeneous cues. For example, [17, 23] show that VLMs can infer location from contextual evidence such as architecture, vehicles, landscapes, and cultural elements, and that chain-of-thought-style reasoning can further improve performance. Similarly, [24, 50] model geolocation as multi-step reasoning over visual, spatial, cultural, and fine-grained location cues. These studies show that compositional reasoning with foundation models can strengthen visual geolocation inference, enabling localization from the kilometer scale down to the block or street level.

Personal identity and social relationship reasoning. Privacy leakage also arises from cues related to personal identity and social relationships, including interpersonal configuration, body language, surrounding people, and broader scene context. Prior work shows that relational attributes emerge from person-pair interactions together with contextual semantics rather than from isolated visible objects [16]. Recent LLM-based methods further strengthen this trend: [18] translates visual content into a textual social narrative and reasons over it to infer social relations, while [20] shows that VLM-agentic frameworks can infer sensitive and abstract user attributes by combining evidence within and across personal images.

Overall, these studies highlight the need to rethink visual privacy as analysis and protection not only against visible identifiers, but also against hidden inferences from contextual reasoning.

2.2 Defenses Against Visual Privacy Threats

Tab. 1 summarizes representative defense solutions against visual privacy threats. We categorize these methods based on the visual privacy threat they address.

Defense against visual geo-localization inference attacks. Prior work mainly protects against specialized CV models that

infer geolocation from images. ADGAN [47], LocGuard [25], and MM-PGD [15] use adversarial learning to modify images and reduce geolocation leakage against CV-based attackers. However, these methods are often optimized for specific model families or feature spaces, limiting their transfer to semantic attackers, such as VLMs. Recent work has begun to consider VLM-based attackers. For example, GeoShield [22] designs adversarial perturbations against VLM geolocation inference through geographical and non-geographical feature disentanglement. Nevertheless, existing defenses still primarily optimize for reducing inference success, rather than giving users explainable, fine-grained control over *which* privacy-relevant clues are removed and *why*. In comparison, VEILLENS designs a reasoning-grounding-editing-verification loop to identify explainable privacy-relevant clues, guide targeted image sanitization, and iteratively verify whether the sensitive attribute remains inferable.

Defense against PII and general privacy leakage. Another line of work focuses on protecting explicit PII and broader visual privacy cues, such as faces, screens, documents, and other identity- or context-revealing content. Prior methods often rely on redaction, selective obfuscation, or task-aware sanitization [32, 40, 44], while newer systems begin to support AI-assisted privacy assessment and mitigation [4, 28]. For example, recent concurrent work CAIAMAR [4] introduces an AI-assisted privacy framework with iterative assessment and mitigation, while Imago Obscura [28] helps users identify and edit privacy risks through a human-centered image privacy co-pilot. GAN- and diffusion-based adversarial editing methods target facial privacy against recognition systems [13, 14], achieving strong identity suppression but remaining limited to face-level threats rather than general compositional inference. Although effective for visible and localized privacy threats, these methods provide limited support for reasoning-based privacy inference, fine-grained clue discovery, or iterative verification.

In comparison, VEILLENS unifies explainable clue discovery, targeted editing, and verification in a single framework, enabling protection against broader threats including PII, social-relationship inference, and geolocation while preserving image usability.

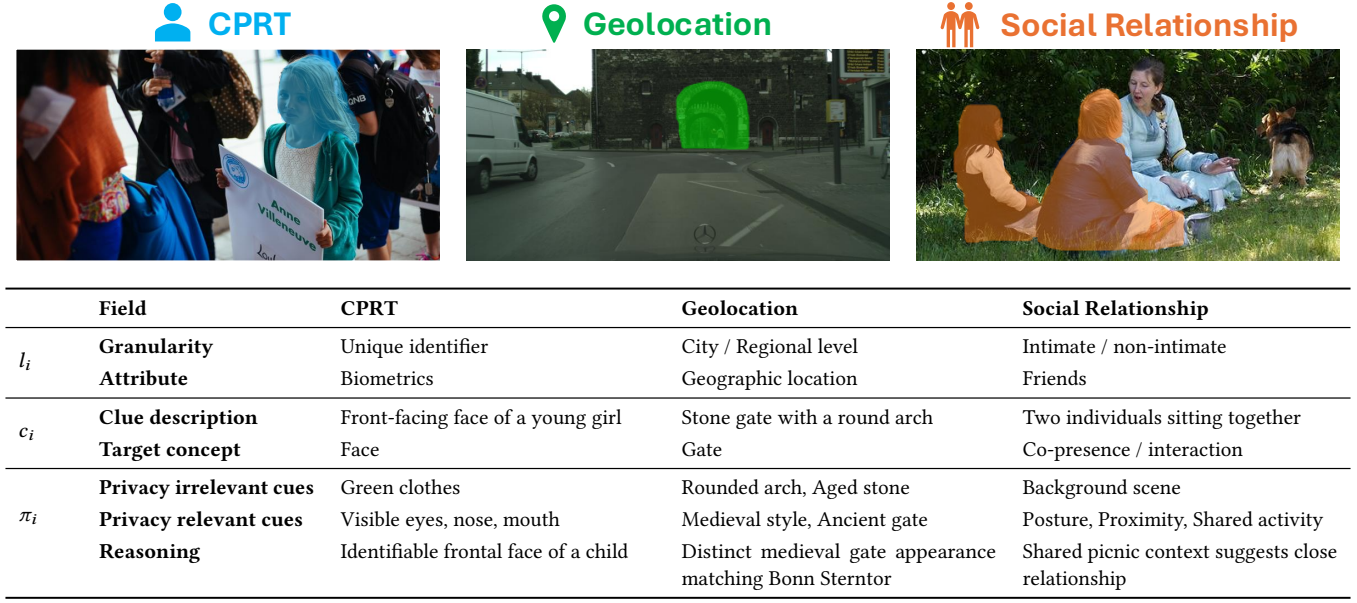


Figure 2: Example clue visualization and its corresponding structured clue representation. The top row shows localized visual evidence from three privacy domains, and the table below maps each highlighted clue to its tuple components (l_i, c_i, π_i)

3 Threat Model and Problem Definition

3.1 Attack Models

We follow the visual privacy attack setting in [55], where a victim captures a photo and shares it through a public or semi-public channel, such as social media, cloud service, online forums, or large messaging groups. After the image is shared, a low-effort, non-expert adversary can obtain the image and use off-the-shelf inference tools, including OCR systems, object detectors, face or scene recognizers, search engines, public map services, or advanced VLMs, to infer sensitive information about the victim. We assume the adversary only has access to the shared image at its platform-provided resolution and quality, without privileged access to private metadata, EXIF data, or other victim-specific context beyond what can be inferred from the image and public information.

Unlike traditional visual privacy attacks that primarily rely on explicit identifiers, we focus on privacy threats that arise from *contextual and compositional visual reasoning*. In these attacks, sensitive information is inferred not from a single obvious object, but from multiple weak cues and their relationships within the scene. Formally, given a shared image x , the adversary aims to produce a sensitive inference y about a victim or scene-associated individual by reasoning over a set of visible cues $\mathcal{E}(x)$.

(1) *Compositional Privacy Risk Taxonomy (CPRT) Attack*: We consider the potential privacy threats by following the CPRT framework proposed by [43], which models visual privacy as a graded and compositional severity-assessment problem.

Besides, we consider two specific representative privacy threats requiring detailed reasoning:

(2) *Geolocation inference attacks*. The adversary aims to infer where the image was captured, i.e., geographic location (precise latitude

and longitude). Such inference may rely on signage fragments, architectural styles, vegetation, road layouts, storefronts, transit systems, weather, indoor layouts, or other region-specific visual evidence. Following recent work on VLM-based geolocation attacks [23], we focus on practical zero-shot semantic reasoning attacks, where the adversary queries publicly accessible VLMs or open-source models to derive and refine location hypotheses from a single shared image. (3) *Social relationship inference attack*. The adversary aims to infer the social relationships among people appearing in the image. Following the visual social relationship recognition setting introduced by [16], we formulate the attack as a pairwise inference problem: given an image I and two people p_i and p_j localized in the image, the adversary predicts a relationship label r_{ij} between them.

The detailed defense granularities against these attacks are discussed in the next section (Sec 3.2).

3.2 Defense Models

Defense setting. VEILLEN is a *user-side* defense framework designed for a *black-box* and *model-agnostic* setting. VEILLEN is applied *before sharing*: it edits the raw image into a defended image that is then uploaded. We assume the VEILLEN pipeline is operated in a trusted execution environment (TEE), that will not itself leak information beyond the defended image. VEILLEN does not rely on knowing which model the adversary uses and assumes completely no access to the attack model parameters. We consider *adaptive adversaries* who may know the existence and strategy of VEILLEN, but can only operate on the defended image. As mentioned in the attack model, we do not aim to defend against (i) privileged platform adversaries, (ii) attackers with additional private context beyond the shared image, or (iii) privacy experts who perform substantial manual investigation using external knowledge.

Privacy protection objective. Given an input image x and a user-specified granularity preference g , VEILLENS outputs a defended image $x' = \text{VEILLENS}(x, g)$ that reduces privacy leakage beyond the level the user is willing to reveal. We define g using a task-specific granularity schema:

(1) *Compositional inference defense*: Level 1 denotes unique identifiers; Level 2 denotes linkage-based identifiers and sensitive personal information; Level 3 denotes aggregation-based identifiers; and Level 4 denotes benign contextual information [43].

(2) *Geolocation inference defense*: We assess geolocation privacy protection across five privacy levels, defined by the attacker’s localization accuracy: *block* (≤ 0.5 km), *street* (≤ 1 km), *neighborhood* (≤ 5 km), *city* (≤ 10 km), and *regional* (≤ 100 km).

(3) *Social relationship inference defense*: We evaluate social relationship privacy protection at two levels of granularity: a 3-way classification, including *intimate*, *non-intimate*, and *no relationship*; and a 6-way classification, including *friends*, *family*, *couples*, *professional*, *commercial*, and *no relationship*.

Such granularity serves as the protection policy for three reasons. First, it distinguishes weak contextual leakage from precise or high-severity inference, providing the pipeline with a priority signal for deciding which clues should be edited, weakened, or preserved. Second, it provides an intuitive user-facing control interface: users specify the level of inference they are willing to reveal, rather than tuning low-level model thresholds or editing parameters. Third, it makes the defense decision explainable by linking each edit to the inferred risk level it aims to suppress. VEILLENS does not guarantee perfect removal of all sensitive information. Instead, it aims to suppress visual evidence that could support inferences more specific or more sensitive than the user’s selected privacy granularity.

Usability and trustworthiness objectives. In addition, VEILLENS aims to satisfy two usability and trustworthiness goals.

First, VEILLENS should *preserve visual fidelity and image utility*. The defended image x' should remain suitable for normal sharing, with minimal visual artifacts, limited unnecessary semantic loss, and preservation of non-sensitive content relevant to the user’s sharing intent.

Second, VEILLENS should provide *explainability*. Rather than outputting only a defended image, VEILLENS highlights the regions, objects, relationships, or contextual cues that contribute to specific privacy risks and explains why they matter. These explanations help users understand how sensitive inferences arise from compositional visual evidence, inspect the privacy–utility trade-off, and choose a protection level that matches their sharing preferences.

4 Methodology

VEILLENS contains four stages: (i) reasoning: identify which visual evidence supports a target privacy inference, (ii) grounding: localize that evidence into editable regions, (iii) editing: transform the image that removes or weakens the inference while preserving overall utility, and (iv) verification: verify whether the desired privacy level has actually been reached after editing.

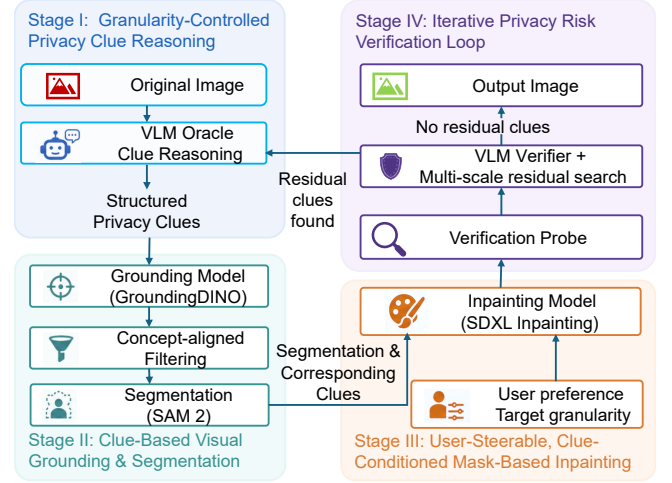


Figure 3: Overview of VEILLENS pipeline. It follows an iterative reasoning–localization–editing–verification workflow.

4.1 Stage I: Granularity-Controlled Privacy Clue Reasoning

Given an input image x , Stage I identifies visual clues that may support privacy-sensitive inference. Rather than detecting a pre-defined set of sensitive object categories, Veillens uses a trusted VLM oracle to produce a finite set of structured privacy clues:

$$F_{\text{VLM}}(x) \mapsto \{e_i = (\ell_i, c_i, \pi_i)\}_{i=1}^k, \quad (1)$$

Each clue e_i denotes a concrete privacy-relevant visual cue:

(1) *Privacy risk category and granularity* (ℓ_i). It characterizes the type and granularity of privacy risk associated with clue e_i . The granularity levels follow the domain-specific definitions in Sec. 3.2. ℓ_i makes explicit how each clue participates in a composite inference structure: some clues act as direct identifiers, while others function as contextual evidence that becomes sensitive only when combined with additional signals.

(2) *Groundable clue description* (c_i). It specifies the visual evidence associated with clue e_i for Stage II grounding (Sec. 4.2). We represent c_i using two components: an *clue description* and a *target concept*. The clue description is a RefCOCO-style referring expression [11]: a concise noun phrase beginning with the target concept and augmented with visible modifiers (e.g., color, shape, material, size) and spatial relations. The target concept corresponds to the leading noun phrase and serves as a semantic anchor for token-aligned candidate filtering (Sec. 4.2).

(3) *Explainability and actionability* (π_i). The field π_i provides an actionable attribution for clue e_i by separating *privacy-relevant cues*, *privacy-irrelevant cues*, and a brief *reasoning statement*. The privacy-relevant elements specify what semantics should be suppressed, while the privacy-irrelevant cues describe visual properties that should be preserved when possible. The reasoning statement explains why these cues support the sensitive inference, helping users judge whether the proposed edit matches their own sharing intent before accepting or rejecting the intervention. This decomposition directly supports Stage III (Sec 4.3).

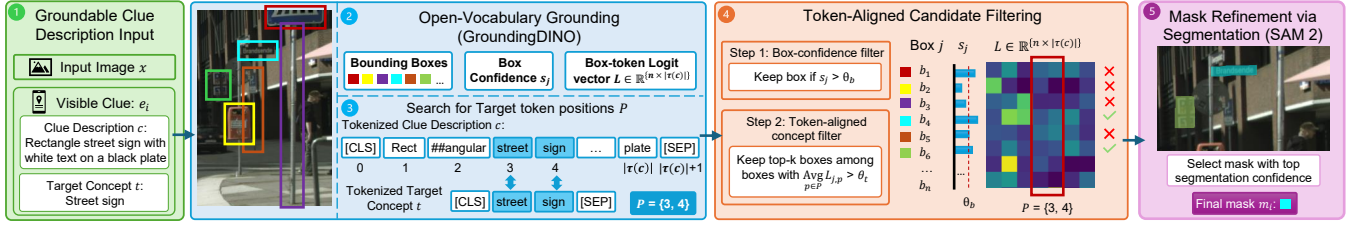


Figure 4: Visual grounding and segmentation pipeline. The target concept is located within the tokenized clue description to obtain the relevant token positions P . GroundingDINO’s bounding-box logit matrix is then filtered by box confidence and concept-specific token activation, retaining boxes that correspond to the intended concept before segmentation refinement.

Fig. 2 shows examples of e_i across different privacy risks. This clue-level formulation differs from existing visual privacy defenses that rely on predefined sensitive labels or isolated visible objects [1]. Such fixed-label formulations are insufficient for reasoning-based privacy threats because leakage is often compositional: individually benign cues can jointly reveal sensitive attributes [43]. For example, geolocation inference may depend on long-tail evidence such as street layout, front-yard design, architecture, vehicles, landscapes, and cultural elements, rather than on a fixed set of sensitive object categories [23, 24].

We adopt a VLM oracle for clue reasoning for two reasons. First, it can reason over heterogeneous and long-tail visual evidence that is difficult to capture with predefined detectors. Second, it can provide natural-language rationales that explain why each clue matters, which supports both user interpretability and downstream editing [28]. Note that, in this step, we assume the oracle VLM is trusted and executed inside TEE, i.e., it can access x to reason about potential privacy leakage evidence but is not allowed to exfiltrate the unprotected image or its sensitive content.

4.2 Stage II: Clue-Based Visual Grounding and Segmentation

Second, VEILLENs grounds the structured clue set $\{e_i = (\ell_i, c_i, \pi_i)\}_{i=1}^k$ produced in Stage I into editable image regions. Next, we introduce the detailed 5 steps for this stage, as shown in Fig. 4.

① Groundable Clue Description Input. A central requirement for Stage II is that the grounding query must be both semantically specific and spatially localizable. Therefore, as introduced in Stage I, VEILLENs represents each clue description c_i as a RefCOCO-style referring expression [11], which provides a concrete instance-level query suitable for open-vocabulary grounding, while preserving enough visual specificity to distinguish among similar objects.

To validate this design choice, we compare RefCOCO-style expressions against two representative alternatives for converting VLM-generated clues into grounding queries. *Free-form descriptions* directly use the VLM-generated explanation of a privacy clue as the grounding query. *Categorical labels* reduce each clue to its primary object, scene, or privacy-attribute noun, such as *license plate*, *street sign*, or *storefront*. This representation follows the attribute-style formulation used in visual privacy datasets such as VISPR [1]. For each generated clue, we run GroundingDINO [21] to produce candidate bounding boxes from the corresponding grounding query. A human annotator then inspects the top-1 confidence bounding

box and counts the result as successful only if the box tightly covers the intended target clue. As shown in Tab. 2, RefCOCO-style expressions achieve the highest grounding success rate. It suggests that RefCOCO-style expressions provide a useful middle ground: they are concise enough to be grounded reliably, while retaining modifiers and spatial context needed to localize the intended instance. Thus, we use RefCOCO-style expressions as our default representation of c_i .

② Open-Vocabulary Grounding. Given an image x and a clue description c_i , we first apply an open-vocabulary grounding operator $\mathcal{G}$:

$$\mathcal{G}(x, c_i) \mapsto (B, L, S) \quad (2)$$

where $B = \{b_j\}_{j=1}^n$ denotes the set of candidate bounding boxes, $L \in \mathbb{R}^{n \times |P(c_i)|}$ is the box-token alignment logit matrix over the tokenized clue $\tau(c_i)$, and $S = \{s_j\}_{j=1}^n \in \mathbb{R}^n$ contains the grounding confidence scores, with s_j corresponding to candidate box b_j . We instantiate $\mathcal{G}$ using GroundingDINO [21], an open-sourced text-conditioned open-vocabulary detector.

③ Targeted Token Position Search. Although open-vocabulary grounding provides candidate boxes, the highest-confidence box does not always correspond to the intended clue. In particular, grounding can be activated by incidental modifiers, nearby objects, or context words in the clue phrase rather than by the target object itself. To improve semantic faithfulness, VEILLENs refines GroundingDINO candidates using the token-level alignment scores produced by the detector. We tokenize both c_i and t_i using the tokenizer in $\mathcal{G}$ and identify the set of token positions $P(t_i, c_i)$ in $\tau(c_i)$ that correspond to t_i by matching token IDs between the target concept and the object description.

④ Token-Aligned Candidate Filtering. We keep boxes that pass both confidence and target-token thresholds:

$$S = \{j : s_j > \theta_b \text{ and } \frac{1}{|P|} \sum_{p \in P} L_{j,p} > \theta_t\}, \quad (3)$$

ranked by mean target-token alignment, with the top- k passed to SAM 2. If P is empty or $S = \emptyset$, we fall back to the confidence-filtered set $\{j : s_j > \theta_b\}$.

⑤ Mask Refinement via Segmentation. The selected boxes provide approximate localization, but box-level regions are too coarse for localized editing (Sec. 4.3). To obtain a pixel-level editing target, we refine each selected box using a segmentation model, i.e., SAM 2 [38]. It produces segmentation mask candidates and confidence score for each selected box. VEILLENs keeps the mask m_i with highest confidence as the final grounded region for clue e_i .

Prompt format	Example output	Grounding success rate
Free-form description	The medieval-looking entrance structure suggests a historic European city center; The traffic signage and surrounding street layout reveal local road conventions.	0.76
Categorical label	Sign; Building.	0.64
RefCOCO-style expression	City gate with dark stone arch; Yellow stop sign beside road.	0.89

Table 2: Comparison of clue-description formats used in Stage I. The grounding success rate measures the fraction of generated clues with correctly produced bounding boxes.

4.3 Stage III: User-Steerable, Clue-Conditioned Mask-Based Generative Inpainting

Given the clue e_i in Stage I and the corresponding m_i in Stage II, Stage III transforms the original image x into a defended image x' through mask-based generative inpainting. The goal is to suppress privacy-revealing visual evidence while preserving the overall appearance, utility, and shareability of the image.

User-steerable privacy editing. As shown in Fig. 1, Stage III provides two complementary forms of user control. First, users can specify a target protection granularity, such as a maximum residual CPRT score, a desired geolocation granularity, or a protected social-relationship category. Because privacy preferences vary across sharing contexts, this allows users to adjust protection strength and choose an appropriate privacy-utility trade-off. Second, users can manually inspect and select individual clues for removal. For each clue e_i , VEILLENS presents its risk category ℓ_i , rationale π_i , and localized mask m_i . This clue-level interface lets users understand why each region is privacy-relevant and accept or reject specific edits. Together, granularity-level control and clue-level selection make editing controllable and explainable rather than a fixed automatic sanitization procedure.

Clue-Conditioned Localized Editing. For each clue e_i selected by the user, VEILLENS performs localized mask-based generative inpainting over the corresponding region m_i . The inpainting operator takes the current image x , the binary mask m_i , a structural control image $q(x)$, and clue-derived textual conditions p_i as input:

$$x' = \mathcal{I}(x, m_i, q(x), p_i^+, p_i^-), \quad (4)$$

where $\mathcal{I}$ denotes a pretrained diffusion-based inpainting model. The structural condition $q(x)$ is a monocular depth map estimated from the original image using a pretrained depth-prediction model [35, 36]. We feed this depth map into ControlNet [52], which injects spatial conditioning into the diffusion model so that the edited region follows the original scene geometry and object layout. The textual conditions p_i^+ and p_i^- specify the intended semantic direction of the local edit. Specifically, privacy-irrelevant cues in π_i form the positive prompt p_i^+ , describing benign visual content that should be retained or reconstructed, while privacy-relevant cues form the negative prompt p_i^- , specifying sensitive semantics that should

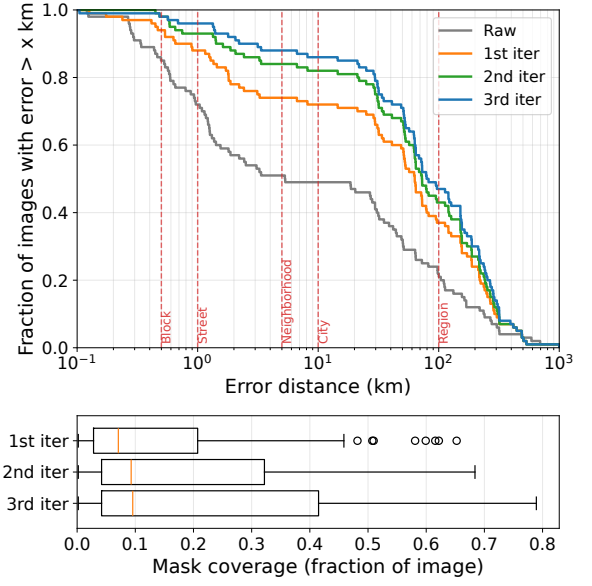


Figure 5: Effect of iterative verification on geolocation privacy. Top: each verification round shifts the error-distance survival curve upward, indicating reduced fine-grained location leakage. Bottom: mask coverage improvement introduced by additional rounds.

not be reproduced. Together, the mask, structural condition, and clue-derived prompts make the edit localized, geometry-aware, and traceable to the original privacy rationale.

Global photo editing and risk feedback. In addition, VEILLENS also supports global photo edits that users may apply for aesthetic or sharing purposes. Such edits include common photographic adjustments and filters that modify the entire image rather than a specific mask. We found that some global effects may reduce residual privacy risk by weakening fine-grained visual cues used by VLM-based attackers. In particular, film grain is a promising candidate because it is a familiar photographic effect and introduces tunable stochastic texture that can reduce the visibility of subtle cues while maintaining a natural image appearance. This allows users to combine localized inpainting with ordinary photo editing while maintaining visibility into the current privacy risk of the shared image.

4.4 Stage IV: Iterative Privacy Risk Verification Loop

A single round of reasoning, grounding, and editing does not necessarily remove all privacy leakage. The first pass usually exposes the most salient evidence, but weaker cues may remain and become sufficient for VLM-based inference once the dominant evidence has been suppressed. This failure mode is especially important for compositional privacy risks, where the sensitive attribute is supported by multiple weak cues rather than one explicit identifier.

To address this, VEILLENS treats protection as an iterative verification process. After each editing round, VEILLENS constructs a

verification probe by masking the clue regions already discovered in previous rounds. The verifier then re-runs the Stage I clue reasoning module on this probe and asks whether the current image still contains residual evidence supporting the initial sensitive prediction. Masking previously handled regions prevents the verifier from repeatedly reporting the same dominant clues and forces the search toward remaining evidence. If residual clues are found, VELLSENS grounds and edits them through the same Stage II–III pipeline.

When the verifier reports residual evidence that is small, or spatially diffuse, VELLSENS applies a multi-scale residual search by dividing image into overlapping patches. Each patch is evaluated with the global prediction as context so that the verifier reports only local evidence that supports the target inference. The loop terminates when no residual clue exceeds the target privacy criterion, or when the predefined iteration budget is reached.

We evaluate the benefit of this iterative design on 100 images under geolocation inference attacks. The target geolocation granularity is set to the city level, so the pipeline focuses on suppressing street- and city-level evidence. As shown in Fig. 5, iterative verification progressively shifts the geolocation error distribution toward larger distances, indicating weaker location leakage. At the street-level threshold (< 1 km), attack success rate drops from 29.0% on raw images to 13.0%, 8.0%, and 5.0% after the first, second, and third rounds, respectively. At the city-level threshold (< 10 km), success similarly decreases from 52.0% to 29.0%, 19.0%, and 15.0%. The first iteration removes the most salient clues and yields the largest gain, while later iterations continue to expose secondary residual cues. The corresponding mask-coverage distribution shows that these gains are achieved with moderate additional editing rather than full-image degradation: the median edited area remains below roughly 10% of the image across rounds. The high-coverage tail mainly arises in street-scene images where residual geolocation evidence is carried by large background structures rather than compact foreground objects, such as building facades, road layout, storefronts, or extended street furniture.

5 Implementation and Default Setup

We implement VELLSENS using commercial off-the-shelf VLM APIs and open-source vision models.

Reasoning and verification VLM calls. All defense-side VLM calls return structured outputs following task-specific schemas. We use GPT-5 as the default trusted VLM oracle for Stage I privacy-clue reasoning and Stage IV residual-risk verification. This oracle is used only inside the defense pipeline and is kept separate from the adversary VLMs used for evaluation. The VLM attackers used for evaluation are described as attacker baselines in Sec. 6. Keeping the defense oracle distinct from the evaluation attackers avoids overestimating protection due to model-specific alignment between the defender and the attacker, and is consistent with our model-agnostic defense design. Importantly, GPT-5 is only the default instantiation of the trusted oracle; VELLSENS treats the VLM oracle as a replaceable component.

Grounding and Editing. For each clue, the grounding stage queries GroundingDINO [21] (SwinT-OGC) at box thresholds of 0.30, retains the top two candidates, and refines them into pixel-level masks via SAM 2.1 [38]; Editing backend is SDXL 1.0 [34] with a depth-ControlNet (controlnet-depth-sdxl-1.0-small) [52]

conditioned on MiDaS DPT-Hybrid [37] depth maps (50 denoising steps, guidance 7.5, ControlNet scale 0.7, strength 0.8); The default pipeline runs verifier twice: Round 1 on the full image; Round 2 with multi scale residual search using overlapping 768×768 crops at 10% overlap.

Execution. All experiments run on an NVIDIA RTX PRO 6000 GPU; We use mixed-precision inference and memory-saving diffusion settings: fp16 weights, TF32 matrix multiplication, attention slicing, and VAE tiling. All images are evaluated at original resolution.

6 Experimental Setup

Our evaluation covers the three categories of privacy risk introduced in our threat model (Sec. 3.2): *compositional inference*, *geolocation inference*, and *social-relationship inference*. Across tasks, VELLSENS follows the same pipeline, with task-specific schemas, metrics, and intervention policies.

General Compositional Privacy Risk.

Metric. We use the Continuous Privacy Risk Score from CPRT [43], which maps privacy-relevant attributes to a severity value in $[0, 1]$. Unlike binary privacy labels, this score captures both explicit identifiers and weaker compositional cues. We report both the post-defense score ($\downarrow$ better) and its reduction relative to the undefended image ($\uparrow$ better). We set the compositional protection target to CPRT Level 2, suppressing both level 1 and 2 identifiers.

Dataset. We evaluate on 500 randomly sampled images from the CPRT dataset [43], which reorganizes VISPR annotations [33] into graded severity levels and continuous risk scores. The subset has a mean severity of 0.48, spanning both low- and high-risk images.

Geolocation Privacy Risk.

Metric. We compute geodesic error ($\uparrow$ better for the defender) using Haversine distance between predicted and ground-truth coordinates. Following the granularity schema defined in our threat model (Sec. 3.2), we bucket geodesic error into five levels: *block* (≤ 0.5 km), *street* (≤ 1 km), *neighborhood* (≤ 5 km), *city* (≤ 10 km), and *regional* (≤ 100 km). Attack success rate at a given level is the fraction of images whose predicted coordinate falls within the corresponding radius of the ground truth ($\downarrow$ better). We set the geolocation target to city level, i.e., within 10 km.

Dataset. We evaluate on 500 images from CITYSCAPES [6], sampled uniformly across 18 European cities. Ground-truth GPS coordinates are extracted from the provided vehicle metadata. Cityscapes is challenging for the defender because its dashcam images contain subtle, distributed urban cues. For cross-dataset robustness, we also build a 200-image MAPILLARY holdout set [27]. We sample 20 global cities with population at least 300,000 and collect 10 non-panoramic street-level images near each city. Images are sampled uniformly within a fixed radius around each city center, subject to Mapillary availability and the non-panoramic constraint. Each image uses its own Mapillary GPS coordinate as ground truth. We report MAPILLARY separately because it differs from CITYSCAPES in capture device, viewpoint, geography, and image distribution.

Social-Relationship Privacy Risk.

Metric. We report two quantities. (i) *Relationship Attack Accuracy* ($\downarrow$ better): the adversary VLM’s relationship-classification accuracy on annotated person pairs, measured before and after defense. We report this under both the original 3-way and 6-way PISC labels

Table 3: Defense baselines used in our evaluation.

Name	Design axis	Core mechanism
RAW	—	Image without defense
RAND-NOISE	Non-semantic global	Replace 10% randomly selected pixels with values from a 51×51 Gaussian-blurred image
OCR-BLUR	Semantic text obfuscation	EAST detection [56] + 31×31 Gaussian blur on text regions
MM-PGD	Adversarial perturbation	PGD [26] attack against a Places365 ensemble, ℓ_∞ $\epsilon=0.5$, $T=20$
GEOSHIELD	CLIP-ensemble adversarial (geo-only)	FGSM against CLIP surrogates with geo-semantic loss [22], square-tiled
VEILLENS	VLM-guided localized editing	Reasoning-guided clue grounding, inpainting, and verification

(Sec. 3.2). (ii) *Attack Reduction* ($\uparrow$ better): the relative decrease in attack accuracy after defense, computed as $(\text{Acc}_{\text{raw}} - \text{Acc}_{\text{def}}) / \text{Acc}_{\text{raw}}$. We set the social-relationship target to no relationship, suppressing evidence for both intimate and non-intimate relationship inference. *Intervention policy.* For social-relationship privacy, we use dataset-provided person boxes and IDs to isolate relationship reasoning from person discovery. During evaluation, VEILLENS reasons about the specified subject pair, and once a relationship-bearing pair is selected, we directly anonymize the face regions within the corresponding boxes. This does not reduce generality, since subject localization can be supplied by standard person detectors.

Dataset. We use PISC [16], which provides person bounding boxes and pairwise relationship labels. We randomly sample 500 images with 1,109 annotated pairs across six categories: friends (307), family-members (109), couple (37), professional (397), commercial (29), and no-relationship (230). During preprocessing, we overlay dataset-provided boxes and IDs so VLM predictions can be matched to the intended subject pair.

Perceptual Quality Metrics. To evaluate utility, we use four perceptual metrics: *SSIM* [46] for structural similarity ($\uparrow$), *LPIPS* [53] for learned perceptual distance ($\downarrow$), *CLIP-IQA* [45] for no-reference quality ($\uparrow$), and *LIQE* [54] for learned image quality evaluation ($\uparrow$).

Attacker Baselines. We evaluate against four frontier adversary VLMs: gemini-3-flash-preview [9], gemini-3-pro-preview [10], gemini-2.5-flash [8], and gpt-5.2 [31]. These models span multiple providers and capability tiers. The attacker set includes models different from, and in some cases newer than, the defense oracle to reduce model-specific evaluation bias.

Defense Baselines. We compare VEILLENS with a no-defense baseline and four representative defense baselines, covering non-semantic perturbation, text obfuscation, adversarial perturbation, and CLIP-based geo-semantic perturbation (used only for geolocation), as shown in Tab. 3. Baselines whose threat model applies only to geolocation are evaluated only on the geolocation task and marked as not applicable for the other tasks.

Anti-Caching Mechanisms. Repeated API queries may be affected by response, prefix, or key-value caching in modern VLM serving stacks, potentially producing correlated outputs or timing artifacts [7, 41]. To reduce such artifacts, we prepend each request with a UUIDv4 nonce, formatted as Request-ID: <uuid>, and instruct the model to ignore it.

7 Quantitative Evaluation Results

Next, we quantitatively evaluate VEILLENS against three representative privacy risks as discussed in the previous section.

7.1 General Compositional Privacy Risk Results

Fig. 7 reports CPRT privacy-risk distributions before and after protection under GPT-5.2 and Gemini-3P attackers.

Raw images expose measurable compositional privacy risk. We evaluate general compositional privacy using the CPRT privacy risk score, where higher scores indicate stronger model-inferred privacy leakage. On the raw images, the mean privacy risk score is 0.3138, showing that many images contain privacy-relevant evidence beyond explicit identifiers. This confirms the central motivation of VEILLENS: ordinary images can expose sensitive information through weak and compositional cues, rather than only through obvious objects such as faces, text, or license plates.

VEILLENS substantially reduces model-inferred compositional privacy risk. Under GPT-5.2, the mean CPRT privacy risk score drops from 0.3138 on raw images to 0.1611 after protection, corresponding to a 48.6% relative risk reduction. Gemini-3P shows a similar trend, reducing the mean score from 0.3138 to 0.1641, a 47.7% relative reduction. This consistency across two frontier attackers suggests that the reduction is not specific to a single model family. As shown in Fig. 7, both post-defense survival curves are consistently below the raw curve, indicating that VEILLENS shifts the full risk distribution toward lower privacy leakage rather than improving only a small subset of images. The more step-like Gemini curve reflects Gemini-3P’s coarser risk-score output granularity, not a smaller evaluation set or a substantively different trend.

VEILLENS is especially effective on high-risk images. Among the 273 high-concern images with ground-truth CPRT score ≥ 0.5 , the mean privacy risk score decreases from 0.5224 to 0.2456 under both GPT-5.2 and Gemini-3P. This corresponds to a 53.0% relative reduction, larger than the reduction on the full set. These results show that VEILLENS reduces high-severity Level 1 and Level 2 leakage, rather than only improving already low-risk cases.

VEILLENS achieves these reductions with localized rather than destructive edits. The median mask coverage is only 0.18% of image pixels, and roughly half of the images are correctly left untouched. The mean mask coverage of 14.3% reflects a smaller subset of images requiring larger interventions. This distribution supports the intended privacy-utility trade-off: VEILLENS applies protection selectively to reasoning-relevant regions instead of relying on blanket blurring, global degradation, or face-only anonymization.

7.2 Geolocation Privacy Risk Results

Tab. 4 reports geolocation attack success rates before and after each defense across four VLM attackers and five distance thresholds.

Table 4: Geolocation attack success rates on the 500-image benchmark. Lower is better; green and yellow mark the best and second-best defenses per column.

Defense	Gemini-3F					Gemini-3P					Gemini-2.5F					GPT-5.2				
	<0.5km	<1km	<5km	<10km	<100km	<0.5km	<1km	<5km	<10km	<100km	<0.5km	<1km	<5km	<10km	<100km	<0.5km	<1km	<5km	<10km	<100km
raw	42.0%	55.6%	71.8%	73.4%	87.0%	38.4%	50.8%	64.8%	67.2%	80.6%	15.8%	29.6%	45.0%	46.4%	69.2%	10.2%	22.0%	33.6%	34.2%	52.0%
rand-noise	25.2%	35.2%	47.6%	48.2%	68.2%	24.0%	32.6%	44.8%	51.8%	64.8%	11.8%	20.4%	31.2%	31.8%	57.2%	3.4%	9.0%	15.4%	16.0%	30.6%
ocr-blur	38.6%	49.4%	65.2%	66.4%	81.4%	33.2%	44.2%	57.2%	62.0%	71.4%	14.4%	28.2%	41.6%	42.8%	64.8%	9.2%	19.2%	30.0%	30.4%	48.0%
mm-pgd	30.8%	42.8%	57.8%	58.8%	76.4%	27.2%	33.8%	50.8%	56.0%	75.4%	14.6%	25.6%	38.0%	39.2%	64.8%	7.6%	15.6%	26.4%	27.0%	44.2%
geoshield	24.8%	35.8%	49.0%	49.8%	65.4%	26.8%	35.4%	51.2%	52.8%	68.0%	9.6%	17.4%	27.6%	28.4%	43.0%	6.6%	15.8%	25.4%	25.8%	39.6%
VEILLENS	12.2%	20.8%	29.6%	30.4%	50.2%	14.0%	21.4%	31.2%	32.0%	51.8%	8.2%	14.8%	21.2%	21.8%	46.6%	1.2%	5.0%	10.8%	11.0%	25.4%

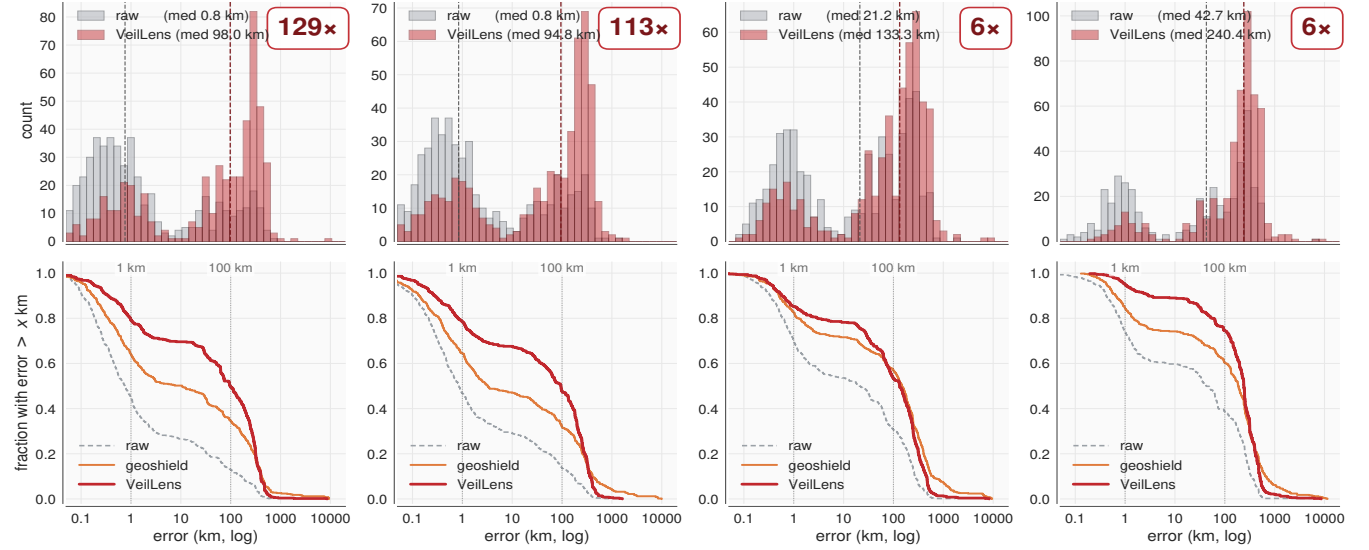


Figure 6: Geolocation evaluation across four VLM evaluators. Top: error distributions before and after protection; VEILLENS shifts errors rightward. Bottom: survival curves; higher curves indicate stronger protection.

Across all evaluated attackers, **VEILLENS substantially reduces fine-grained geolocation success**. Undefended images expose substantial geolocation risk under recent strong VLMs. At the most privacy-sensitive 0.5 km threshold, Gemini-3F achieves a 42.0% success rate on raw images, compared with 15.8% for Gemini-2.5F and 10.2% for GPT-5.2. This suggests that more capable VLMs can localize a large fraction of images to near block-level precision, making geolocation leakage an increasingly severe threat. **VEILLENS** significantly reduces this risk. Even under Gemini-3P, for which the raw 0.5 km success rate is 38.4%, **VEILLENS** reduces the success rate to 14.0%, corresponding to a 2.74 $\times$ decrease.

VEILLENS provides the strongest protection across all evaluated defense baselines. For Gemini-3F at the 0.5 km threshold, the strongest prior baseline is **GEOSHIELD** [22] at 24.8%. **VEILLENS** further reduces success to 12.2%, a 2.03 $\times$ lower attack success rate than the strongest prior defense.

VEILLENS provides consistent protection across distance thresholds and is especially effective against high-risk images. To assess whether **VEILLENS** improves geolocation privacy beyond a few selected thresholds, Fig. 6 reports error distributions

and survival curves for **RAW**, **GEOSHIELD**, and **VEILLENS**. For threshold x , the y-axis of the survival curve denotes the fraction of images with geolocation error exceeding x , so higher curves indicate stronger protection. Across all four evaluators, **VEILLENS** remains above both **RAW** and **GEOSHIELD** over most of the threshold range, showing that the benefit reflects a broad shift in the full error distribution rather than improvement at only a few discrete cutoffs. The advantage is particularly clear in the privacy-sensitive regime below 10 km, where successful localization can reveal block-, street-, or district-level information. For Gemini-2.5F, the improvement over **GEOSHIELD** is more modest because the raw geolocation risk is already lower for this weaker attacker.

VEILLENS is general to different image datasets. We further evaluate **VEILLENS** on a 200-image **MAPILLARY** dataset to test whether its geolocation protection transfers beyond the **CITYSCAPES** distribution. Compared with **CITYSCAPES**, **MAPILLARY** contains more diverse geographies, camera sources, viewpoints, and scene compositions, making it a stronger test of dataset robustness. **VEILLENS** substantially reduces fine-grained geolocation success on this set. At the most privacy-sensitive 0.5 km threshold, Gemini-3F achieves a 14.6% success rate on raw images, while **VEILLENS** reduces the

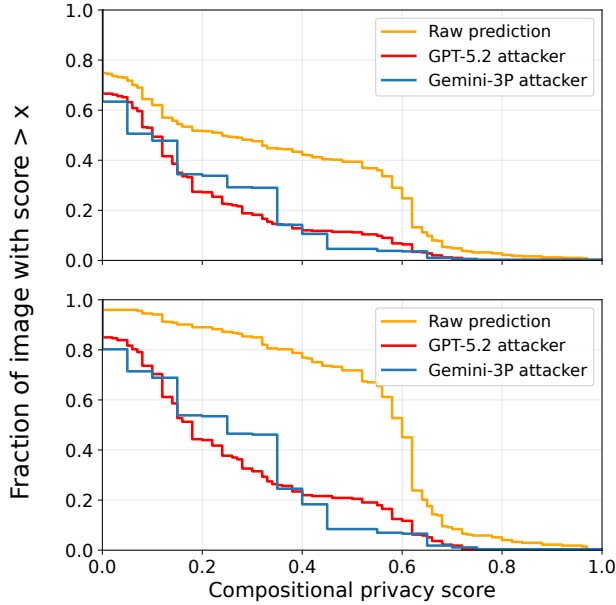


Figure 7: CPRT privacy-risk score distributions before and after protection; lower is better. Top: all images. Bottom: high-risk subset with ground-truth CPRT score ≥ 0.5 .

success rate to 4.7%, corresponding to a $3.11\times$ decrease. In comparison, GEOSHIELD only reduces the success rate to 10.9%. The same trend holds at coarser thresholds: VEILLENS reduces 1 km success from 23.2% to 14.8% and 10 km success from 41.6% to 23.0%. These results suggest that VEILLENS reduces geolocation-inference risk by suppressing transferable visual evidence.

7.3 Social-Relationship Privacy Risk

Social relationships remain inferable from raw images. We evaluate social-relationship inference attack on the PISC dataset [16] using the annotated person pairs as evaluation targets. The task measures whether a VLM can infer the relationship between a specified pair from visual context. As shown in Tab. 6, raw six-way relationship accuracy ranges from 67.5% to 73.1% across four frontier VLM attackers, while raw three-way grouped accuracy ranges from 71.4% to 78.3%. These results show that social-relationship leakage is measurable: even without identity metadata, VLMs can infer relationship categories from visible subject appearance, spatial arrangement, activity, and scene context.

VEILLENS substantially suppresses residual relationship inference. After VEILLENS anonymizes the relationship-bearing subjects, six-way attack accuracy drops to 4.1%–5.8%, and three-way grouped accuracy drops to 4.3%–6.1%. This corresponds to a 92.0%–93.9% relative reduction under the six-way labels and a 92.2%–94.0% relative reduction under the three-way grouping. The reduction is consistent across all evaluated attackers, indicating that the defended images no longer preserve enough visible subject information for reliable relationship classification. The remaining predictions are primarily no-relationship cases, which fall outside the protected relationship target.

7.4 Visual Quality After Protection

We report perceptual quality on the geolocation benchmark because it is the setting where all defenses edit the same image distribution and can be directly compared. For CPRT, many images are unchanged or edited only in small regions, making image-level quality scores less informative; for social relationships, the intervention is deterministic face anonymization rather than general clue-conditioned editing.

VEILLENS provides a stronger privacy-aesthetics balance than prior defenses. Practical deployment requires protected images to remain visually acceptable for everyday sharing. Tab. 5 compares perceptual quality across defenses using image quality metrics. Regional or weak perturbation baselines preserve image quality but provide limited protection: OCR-BLUR maintains high visual quality by blurring only detected text regions, while MM-PGD remains visually subtle due to its small adversarial perturbations; however, both are less effective at suppressing fine-grained geolocation inference (Tab. 4). In contrast, the strongest prior privacy baseline, GEOSHIELD, reduces geolocation leakage more effectively but introduces substantial visual distortion. Compared with GEOSHIELD, VEILLENS achieves substantially better perceptual quality, improving SSIM from 0.63 to 0.79, reducing LPIPS from 0.39 to 0.22, and increasing LIQE from 49.2 to 58.6. These results show that VEILLENS achieves stronger privacy protection than visually conservative baselines while preserving substantially better visual quality than aggressive prior defenses.

Table 5: Perceptual quality on the 500-image geolocation benchmark. Higher is better except LPIPS.

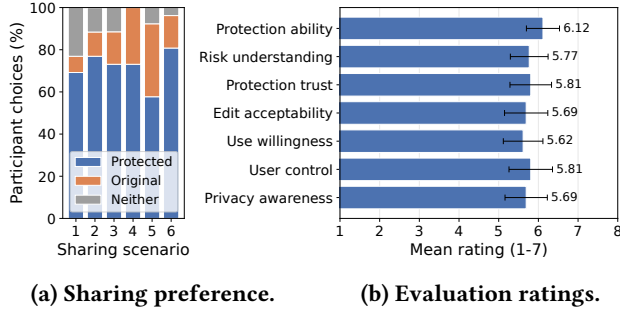
Defense	SSIM $\uparrow$	LPIPS $\downarrow$	CLIP-IQA $\uparrow$	LIQE $\uparrow$
RAND-NOISE	0.74	0.31	0.71	54.3
OCR-BLUR	0.88	0.12	0.76	60.8
MM-PGD	0.85	0.18	0.72	56.1
GEOSHIELD	0.63	0.39	0.65	49.2
VEILLENS	0.79	0.22	0.74	58.6

8 User Study

We conducted an IRB-exempt formative user study to show whether users understand the risks or accept the protected outputs in realistic sharing scenarios. $N = 25$ participants are recruited to evaluate three human-centered questions: (1) whether VEILLENS surfaces overlooked privacy risks, (2) whether its clue-level explanations are understandable, and (3) whether the edited images remain acceptable for sharing. Participants reviewed eight examples covering six location scenarios, one face/identity case, and one document case. For each example, they saw the original image, the protected image, the original and residual inference, a textual rationale, and a visual highlight overlay marking the modified regions. They then completed seven-point Likert ratings and a semi-structured interview. **VEILLENS makes latent privacy risks salient to users.** Participants had moderate baseline privacy concern ($M = 4.49$, $SD = 1.52$) and moderate prior awareness that AI systems can infer private information from photos ($M = 4.64$, $SD = 1.52$). After reviewing VEILLENS’s examples, they rated the original location images as

Table 6: Social-relationship attack accuracy before and after VEILLENS protection across four VLM attackers. Lower is better. Reduction is computed as $(Acc_{raw} - Acc_{def})/Acc_{raw}$. Higher is better.

Attack VLM	6-way Raw Acc. ↓	6-way After VEILLENS ↓	6-way Reduction ↑	3-way Raw Acc. ↓	3-way After VEILLENS ↓	3-way Reduction ↑
Gemini-3 Flash Preview	68.3%	4.5%	93.4%	72.6%	4.7%	93.5%
Gemini-3 Pro Preview	73.1%	5.6%	92.3%	77.8%	6.1%	92.2%
Gemini-2.5 Flash	67.5%	4.1%	93.9%	71.4%	4.3%	94.0%
GPT-5.2	72.9%	5.8%	92.0%	78.3%	5.9%	92.5%



Context	Home	Public	Travel
# Participants	18	11	7
Context	Photo w/ People	Photo w/ Docs	Other
# Participants	13	13	1

(c) Where users would deploy VEILLENS.

Figure 8: User study results ($N = 25$). (a) Protected images were selected in 106/150 preference responses (70.7%) across the six sharing scenarios. (b) All seven rating dimensions were above the neutral midpoint (4); protection ability was rated highest, while use willingness was lowest. Error bars denote 95% confidence intervals. (c) Top deployment interests; participants could select multiple contexts.

exposing substantial location information ($M = 5.43$, $SD = 1.53$) and reported being surprised by the extent of leakage ($M = 4.75$, $SD = 1.80$). This suggests that VEILLENS helped participants recognize privacy risks that are not obvious from explicit identifiers alone, especially in ordinary non-landmark images.

Clue-level explanations are useful, but users want control. Participants generally found VEILLENS’s visual highlights and rationales understandable: the overlays helped them see why an image was sensitive (Privacy awareness, $M = 5.69$, $SD = 1.35$), they understood why the system modified the image (Risk Understanding, $M = 5.77$, $SD = 1.34$), and explanations supported trust in the edits (Protection trust, $M = 5.81$, $SD = 1.38$). Participants also wanted the ability to refine or reject edits, rating override capability highly (User control, $M = 5.62$, $SD = 1.36$). This indicates that explanation alone is insufficient for deployment; users also need control over edit strength, boundaries, and final appearance.

Overall, participants strongly agreed that VEILLENS addresses a real privacy problem (Protection ability, $M = 6.12$, $SD = 1.04$), and

52% (13/25) rated their willingness to use the current prototype at least 6 out of 7. Participants were most interested in near-home photos (18/25), photos with people (13/25), and personal documents (13/25), suggesting that VEILLENS is most valued when images are tied to persistent identity, routines, or sensitive records.

9 Discussion and Limitations

First, VEILLENS relies on a VLM oracle to identify privacy-relevant clues. As a result, some cues may be missed and remain unedited, particularly for compositional privacy risks that arise from multiple weak or indirect visual signals. Although the verification loop helps reduce this risk by re-evaluating the edited image, it cannot guarantee exhaustive cue discovery.

Second, our current implementation uses API-based frontier VLMs for clue discovery and verification. This design allows VEILLENS to leverage strong reasoning capabilities, but it may raise privacy and deployment concerns when images must be sent to external services. A more privacy-preserving deployment could instead use a self-hosted model within a trusted execution environment or another locally controlled inference infrastructure.

Third, generative editing may introduce visual artifacts or semantic drift, especially when privacy-sensitive evidence is tightly entangled with content that users wish to preserve. Additional verification and editing rounds can further reduce residual leakage, but they may also increase runtime, cost, and the visual footprint of the defense. A stronger deployment should therefore incorporate a stopping rule that jointly considers residual privacy risk, perceptual quality, user preferences, sharing context, and overall cost.

Finally, future work should extend VEILLENS from single-image protection to video and longitudinal privacy protection. A practical privacy assistant should reason not only about the current image, but also about user context, sharing intent, and previously shared content. The same visual cue may be harmless in isolation, yet become sensitive when linked across posts to a home, workplace, routine, social relationship, or identity.

10 Conclusion

We presented VEILLENS, an end-to-end framework for protecting visual privacy against VLM-based inference attacks. VEILLENS exposes a growing gap between what users perceive as visually safe and what modern VLMs can infer from contextual and compositional cues. By identifying privacy-relevant clues, localizing editable regions, applying targeted generative edits, and verifying residual risk, VEILLENS takes a step toward making visual privacy risks explainable, controllable, and verifiable before images are shared.

References

- [1] Sara Abdulaziz, Giacomo D'Amicantonio, and Egor Bondarev. 2025. Evaluation of Human Visual Privacy Protection: A Three-Dimensional Framework and Benchmark Dataset. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCV Workshops)*. <https://doi.org/10.48550/arXiv.2507.13981> arXiv:2507.13981 Accepted at ICCV 2025 Workshop CV4BIOM.
- [2] Alessandro Acquisti, Laura Brandimarte, and George Loewenstein. 2015. Privacy and Human Behavior in the Age of Information. *Science* 347, 6221 (2015), 509–514. <https://doi.org/10.1126/science.aaa1465>
- [3] Shane Ahern, Dean Eckles, Nathaniel S. Good, Simon King, Mor Naaman, and Rahul Nair. 2007. Over-Exposed? Privacy Patterns and Considerations in Online and Mobile Photo Sharing. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '07). Association for Computing Machinery, New York, NY, USA, 357–366. <https://doi.org/10.1145/1240624.1240683>
- [4] Robert Aufschläger, Jakob Folz, Gautam Savaliya, Manjitha D. Vidanalage, Michael Heigl, and Martin Schramm. 2026. Towards Context-Aware Image Anonymization with Multi-Agent Reasoning. *arXiv preprint arXiv:2603.27817* (2026). <https://doi.org/10.48550/arXiv.2603.27817> arXiv:2603.27817
- [5] Pithayuth Charnsathikul, Almajd Zunquti, Gale Lucas, and Jelena Mirkovic. 2025. Navigating Social Media Privacy: Awareness, Preferences, and Discoverability. *Proceedings on Privacy Enhancing Technologies* 2025, 4 (2025), 620–638. <https://doi.org/10.56553/popets-2025-0148>
- [6] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. 2016. The Cityscapes Dataset for Semantic Urban Scene Understanding. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 3213–3223.
- [7] Zibo Gao, Junjie Hu, Feng Guo, Yixin Zhang, Yinglong Han, Siyuan Liu, Haiyang Li, and Zhiqiang Lv. 2025. I Know What You Said: Unveiling Hardware Cache Side-Channels in Local Large Language Model Inference. In *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, 1649–1668. <https://www.usenix.org/conference/usenixsecurity25/presentation/gao-zibo> Highlights hardware-level cache determinism risks.
- [8] Gemini Team, Google. 2025. Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities. *arXiv preprint arXiv:2507.06261* (2025). arXiv:2507.06261 <https://arxiv.org/abs/2507.06261>
- [9] Google DeepMind. 2025. *Gemini 3 Flash Model Card*. Technical Report. Google DeepMind. <https://storage.googleapis.com/deepmind-media/ModelCards/Gemini-3-Flash-Model-Card.pdf>
- [10] Google DeepMind. 2025. *Gemini 3 Pro Frontier Safety Framework Report*. Technical Report. Google DeepMind. https://storage.googleapis.com/deepmind-media/gemini/gemini_3_pro_fsfs_report.pdf
- [11] Sahar Kazemzadeh, Vicente Ordonez, Mark Matten, and Tamara Berg. 2014. RefertGame: Referring to Objects in Photographs of Natural Scenes. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Doha, Qatar, 787–798. <https://doi.org/10.3115/v1/D14-1086>
- [12] Spyros Kokolakis. 2017. Privacy Attitudes and Privacy Behaviour: A Review of Current Research on the Privacy Paradox Phenomenon. *Computers & Security* 64 (2017), 122–134. <https://doi.org/10.1016/j.cose.2015.07.002>
- [13] Minh-Ha Le and Niklas Carlsson. 2024. StyleAdv: A Usable Privacy Framework Against Facial Recognition with Adversarial Image Editing. *Proceedings on Privacy Enhancing Technologies* 2024, 2 (2024), 106–123. <https://doi.org/10.56553/popets-2024-0043>
- [14] Minh-Ha Le and Niklas Carlsson. 2025. DiffPrivate: Facial Privacy Protection with Diffusion Models. *Proceedings on Privacy Enhancing Technologies* 2025, 2 (2025), 54–70. <https://doi.org/10.56553/popets-2025-0049>
- [15] Trung-Nghia Le, Ta Gu, Huy H. Nguyen, and Isao Echizen. 2022. Rethinking Adversarial Examples for Location Privacy Protection. In *Proceedings of the IEEE International Workshop on Information Forensics and Security*. <https://doi.org/10.48550/arXiv.2206.14020> arXiv:2206.14020
- [16] Junnan Li, Yongkang Wong, Qi Zhao, and Mohan S. Kankanhalli. 2017. Dual-Glance Model for Deciphering Social Relationships. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. <https://doi.org/10.48550/arXiv.1708.00634> arXiv:1708.00634
- [17] Ling Li, Yao Zhou, Yuxuan Liang, Fugee Tsung, and Jiaheng Wei. 2025. Recognition through Reasoning: Reinforcing Image Geo-Localization with Large Vision-Language Models. In *Advances in Neural Information Processing Systems*.
- [18] Wanhua Li, Zibin Meng, Jiawei Zhou, Donglai Wei, Chuang Gan, and Hanspeter Pfister. 2024. SocialGPT: Prompting LLMs for Social Relation Reasoning via Greedy Segment Optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*. <https://doi.org/10.48550/arXiv.2410.21411> arXiv:2410.21411
- [19] Dongyu Liu, Xuhong Wang, Cen Chen, Yanhao Wang, Shengyue Yao, and Yilun Lin. 2025. SVIA: A Street View Image Anonymization Framework for Self-Driving Applications. In *Proceedings of the IEEE Intelligent Transportation Systems Conference*. <https://doi.org/10.48550/arXiv.2501.09393> arXiv:2501.09393
- [20] Feiran Liu, Yuzhe Zhang, Xinyi Huang, Yanan Peng, Xinfeng Li, Lixu Wang, Yutong Shen, Ranjie Duan, Simeng Qin, Xiaojun Jia, Qingsong Wen, and Wei Dong. 2025. The Eye of Sherlock Holmes: Uncovering User Private Attribute Profiling via Vision-Language Model Agentic Framework. *arXiv preprint arXiv:2505.19139* (2025). <https://doi.org/10.48550/arXiv.2505.19139> arXiv:2505.19139 In Proceedings of the ACM International Conference on Multimedia.
- [21] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. 2024. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. In *Proceedings of the European Conference on Computer Vision (ECCV)*. Springer, 38–55.
- [22] Xinwei Liu, Xiaojun Jia, Yuan Xun, Simeng Qin, and Xiaochun Cao. 2026. GeoShield: Safeguarding Geolocation Privacy from Vision-Language Models via Adversarial Perturbations. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [23] Yi Liu, Gelei Deng, Junchen Ding, Yuekang Li, Tianwei Zhang, Weisong Sun, Yaowen Zheng, and Jingquan Ge. 2025. Mission: Impossible—Image-Based Geolocation with Large Vision Language Models. *Proceedings on Privacy Enhancing Technologies* 2025, 4 (2025), 410–428. <https://doi.org/10.56553/popets-2025-0137>
- [24] Weidi Luo, Tianyu Lu, Qiming Zhang, Xiaogeng Liu, Bin Hu, Yue Zhao, Jieyu Zhao, Song Gao, Patrick McDaniel, Zhen Xiang, and Chaowei Xiao. 2025. Doxing via the Lens: Revealing Location-related Privacy Leakage on Multi-modal Large Reasoning Models. In *International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.2504.19373> arXiv:2504.19373
- [25] Wanlun Ma, Derui Wang, Chao Chen, Sheng Wen, Gaolei Fei, and Yang Xi-ang. 2024. LocGuard: A Location Privacy Defender for Image Sharing. *IEEE Transactions on Dependable and Secure Computing* 21, 6 (2024), 5526–5537. <https://doi.org/10.1109/TDSC.2024.3376929>
- [26] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. 2018. Towards Deep Learning Models Resistant to Adversarial Attacks. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- [27] Mapillary. 2026. Mapillary API Documentation. <https://www.mapillary.com/developer/api-documentation>. Accessed: 2026-05-18.
- [28] Kyzyl Monteiro, Yuchen Wu, and Sauvik Das. 2025. Imago Obscura: An Image Privacy AI Co-Pilot to Enable Identification and Mitigation of Risks. *arXiv preprint arXiv:2505.20916* (2025). <https://doi.org/10.48550/arXiv.2505.20916> arXiv:2505.20916 In Proceedings of the ACM Symposium on User Interface Software and Technology.
- [29] OpenAI. 2024. *GPT-4o System Card*. Technical Report. OpenAI. arXiv:2410.21276 <https://arxiv.org/abs/2410.21276>
- [30] OpenAI. 2025. *GPT-5 System Card*. Technical Report. OpenAI. arXiv:2601.03267 <https://cdn.openai.com/gpt-5-system-card.pdf>
- [31] OpenAI. 2025. *Update to GPT-5 System Card: GPT-5.2*. Technical Report. OpenAI. https://cdn.openai.com/pdf/3a4153c8-c748-4b71-8e31-aecbde944f8d/oi_5_2_system-card.pdf
- [32] Tribhuvanesh Orekondy, Mario Fritz, and Bernt Schiele. 2018. Connecting Pixels to Privacy and Utility: Automatic Redaction of Private Information in Images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8466–8475. <https://doi.org/10.1109/CVPR.2018.00883>
- [33] Tribhuvanesh Orekondy, Bernt Schiele, and Mario Fritz. 2017. Towards a Visual Privacy Advisor: Understanding and Predicting Privacy Risks in Images. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. <https://doi.org/10.48550/arXiv.1703.10660> arXiv:1703.10660
- [34] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2024. SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- [35] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. 2021. Vision Transformers for Dense Prediction. arXiv:2103.13413 <https://arxiv.org/abs/2103.13413>
- [36] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. 2022. Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-Shot Cross-Dataset Transfer. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 3 (2022), 1623–1637.
- [37] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. 2022. Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-Shot Cross-Dataset Transfer. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 3 (2022), 1623–1637.
- [38] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. 2024. SAM 2: Segment Anything in Images and Videos. *arXiv preprint arXiv:2408.00714* (2024).
- [39] Stefan Schubert, Peer Neubert, Sourav Garg, Michael Milford, and Tobias Fischer. 2023. Visual Place Recognition: A Tutorial. *IEEE Robotics & Automation Magazine* 31, 3 (2023), 139–153.

- [40] Jiayu Shu, Rui Zheng, and Pan Hui. 2018. Cardea: Context-Aware Visual Privacy Protection for Photo Taking and Sharing. In *Proceedings of the 9th ACM Multimedia Systems Conference* (Amsterdam, Netherlands) (MMSys '18). Association for Computing Machinery, New York, NY, USA, 304–315. <https://doi.org/10.1145/3204949.3204973>
- [41] Linke Song, Zixuan Pang, Wenhao Wang, Zihao Wang, XiaoFeng Wang, Hongbo Chen, Wei Song, Yier Jin, Dan Meng, and Rui Hou. 2024. The Early Bird Catches the Leak: Unveiling Timing Side Channels in LLM Serving Systems. *IEEE Transactions on Information Forensics and Security* (2024). arXiv:2409.20002 <https://arxiv.org/abs/2409.20002> Demonstrates how shared caches induce TTFT leakage and enable prompt inference.
- [42] Efthymios Tsaprazlis, Tiantian Feng, Anil Ramakrishna, Rahul Gupta, and Shrikanth Narayanan. 2025. Assessing Visual Privacy Risks in Multimodal AI: A Novel Taxonomy-Grounded Evaluation of Vision-Language Models. <https://doi.org/10.48550/arXiv.2509.23827> arXiv:2509.23827
- [43] Efthymios Tsaprazlis, Tiantian Feng, Anil Ramakrishna, Sai Praneeth Karimireddy, Rahul Gupta, and Shrikanth Narayanan. 2026. Rethinking Visual Privacy: A Compositional Privacy Risk Framework for Severity Assessment with VLMs. <https://doi.org/10.48550/arXiv.2603.21573> arXiv:2603.21573
- [44] Hari Venugopalan, Zainul Abi Din, Trevor Carpenter, Jason Lowe-Power, Samuel T. King, and Zubair Shafiq. 2023. Aragorn: A Privacy-Enhancing System for Mobile Cameras. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4, Article 181 (2023), 31 pages. <https://doi.org/10.1145/3631406>
- [45] Jianyi Wang, Kelvin C. K. Chan, and Chen Change Loy. 2023. Exploring CLIP for Assessing the Look and Feel of Images. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI, Vol. 37)*. 2555–2563.
- [46] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. 2004. Image Quality Assessment: From Error Visibility to Structural Similarity. *IEEE Transactions on Image Processing* 13, 4 (2004), 600–612.
- [47] Zuobin Xiong, Zhipeng Cai, Qilong Han, Arwa Alrawais, and Wei Li. 2021. ADGAN: Protect Your Location Privacy in Camera Data of Auto-Driving Vehicles. *IEEE Transactions on Industrial Informatics* 17, 9 (2021), 6200–6210. <https://doi.org/10.1109/TII.2020.3032352>
- [48] Alessio Xompero and Andrea Cavallaro. 2025. Learning Privacy from Visual Entities. *Proceedings on Privacy Enhancing Technologies* 2025, 3 (2025), 261–281. <https://doi.org/10.56553/popets-2025-0098>
- [49] Anran Xu, Zhongyi Zhou, Kakeru Miyazaki, Ryo Yoshikawa, Simo Hosio, and Koji Yatani. 2023. DIPA2: An Image Dataset with Cross-Cultural Privacy Perception Annotations. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4 (2023), 1–30. <https://doi.org/10.1145/3631439>
- [50] Sahiti Yerramilli, Nilay Pande, Rynaa Grover, and Jayant Sravan Tamarapalli. 2025. GeoChain: Multimodal Chain-of-Thought for Geographic Reasoning. *arXiv preprint arXiv:2506.00785* (2025). <https://doi.org/10.48550/arXiv.2506.00785> arXiv:2506.00785 In Findings of the Association for Computational Linguistics: EMNLP 2025.
- [51] Guangsheng Zhang, Bo Liu, Tianqing Zhu, Andi Zhou, and Wanlei Zhou. 2023. Privacy Intelligence: A Survey on Image Privacy in Online Social Networks. *Comput. Surveys* 55, 6, Article 113 (2023), 35 pages. <https://doi.org/10.1145/3547299>
- [52] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023. Adding Conditional Control to Text-to-Image Diffusion Models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 3836–3847.
- [53] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 586–595.
- [54] Weixia Zhang, Guangtao Zhai, Ying Wei, Xiaokang Yang, and Kede Ma. 2023. Blind Image Quality Assessment via Vision-Language Correspondence: A Multi-task Learning Perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14071–14081.
- [55] Ruoyu Zhao, Yushu Zhang, Tao Wang, Wenying Wen, Yong Xiang, and Xiaochun Cao. 2023. Visual Content Privacy Protection: A Survey. *Comput. Surveys* (2023). <https://doi.org/10.48550/arXiv.2303.16552> arXiv:2303.16552
- [56] Xinyu Zhou, Cong Yao, He Wen, Yuzhi Wang, Shuchang Zhou, Weiran He, and Jiajun Liang. 2017. EAST: An Efficient and Accurate Scene Text Detector. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2642–2651. <https://arxiv.org/abs/1704.03155>